Lab M8.01 – Designing a Data Lake with Raw / Silver / Gold Zones

Step 1: Data Lake Fundamentals

What is a data lake?

Data lake is a centralized storage system fed with raw data at large scale.

Data lake purpose is to store in a fast way (schema-read) all types of data (structured, semi-structured and non-structured data) that we can clean and validate afterwards. It gives priority to the availability rather than consistency.

Compared to old data storage, data lakes can store any data format, schema on read and raw data is kept after processing it. Data lake has a medallion structure: bronze (raw data), silver (cleaned data), gold (curated data).

Why not store everything in one folder?

Without layers, data lakes would become data swamps. Tons of data that is not possible to extract a meaning out of it.

What problems do zones solve?

It solves the problem of having mixed raw and clean data together, the lack of traceability and the trust on your data. It becomes easy to reprocess as raw data is always kept and each layer has a purpose.

Step 2: Zone Responsibilities

Raw / Bronze Zone

- Data format: all types of original formats
- Level of validation: no validation, schema on read
- Intended consumers: engineers
- Purpose: store raw data quickly
- Characteristics: no business logic, no transformations, no deletions, append-only, must be preserved (immutable)

Cleaned / Silver Zone

- Data format: standardized
- Level of validation: technically correct
- Intended consumers: engineers, data scientists
- Purpose: cleaning and validation
- Characteristics: re-processable, reusable, stable schemas

Curated / Gold Zone

- Data format: related to business purpose
- Level of validation: business correct
- Intended consumers: Analyst, stakeholders, BI tools

- Purpose: built specifically for business decisions
- Characteristics: re-processable, optimized, consumer-ready

Step 3: Folder Structure Design

Sales data

- How is it organized in Raw?
- How is it organized in Silver?
- How is it organized in Gold?
- What partitioning strategy is used?

Customer data

- How is it organized in Raw?
- How is it organized in Silver?
- How is it organized in Gold?
- What partitioning strategy is used?

Product data

- How is it organized in Raw?
- How is it organized in Silver?
- How is it organized in Gold?
- What partitioning strategy is used?

```
/data-lake
 /bronze
  /source=API
   /year=2024
    /month=03
     /day=01
      /sales_2024_03_01.json
  /source=CRM
   /year=2024
    /month=03
     /day=01
      /customers_2024_03_01.csv
  /source=ERP
   /year=2024
```

```
    /month=03
      /day=01
        /products_2024_03_01.json
/silver
  /sales
    /year=2024
      /month=03
        /day=01
          /sales_cleaned_2024_03_01.parquet
  /customers
    /year=2024
      /month=03
        /day=01
          /customers_cleaned_2024_03_01.parquet
  /products
    /year=2024
      /month=03
        /day=01
          /products_cleaned_2024_03_01.parquet
/gold
  /sales_summary
    /year=2024
      /month=03
        /sales_summary_2024_03.parquet
  /customer_360
    /year=2024
      /month=03
        /customer_360_2024_03.parquet
```

Step 4: Data Flow Explanation

All kinds of data are stored as raw in the bronze layer with no validation in the file server. Then transformations occur in the silver layer, where cleaning and validation is performed. Format is standardized to parquet and schema is created. Finally, pre-aggregations and pre-joins are created in the gold model according to business purposes ready to be used by final users.